

CN:

- Extensive discussion of depression and symptoms of depression
- Brief mention of suicide

Cases in evolutionary psychiatry



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ANT215



Some key reminders

- Feedback surveys:
 - For TAs (in person, today!)
 - For me/course (online)
- Important dates:
 - **Full RPD** due on **Tuesday before 11.59PM**. *Late work strictly not marked.*
 - **Final exam** on **Thursday 11th December**. Official start time: 9.15. **Be there at 9AM sharp!** *No retakes.*
 - Location: **Atwood 215**
 - Review session next Tuesday: bring questions!

Can a chimpanzee get depressed?

Stand up for **yes**

Stay seated for **no**

Can a bacterium get depressed?

Stand up for **yes**

Sit down for **no**

Depressive symptoms

≥5 symptoms during the same two week period that are a change from previous functioning; depressed mood and/or loss of interest/pleasure must be present; exclude symptoms clearly attributable to another medical condition

**Depressed mood**

Most of the day, nearly every day; may be subjective (e.g. feels sad, empty, hopeless) or observed by others (e.g. appears tearful); in children and adolescents, can be irritable mood

**Loss of interest/pleasure**

Markedly diminished interest/pleasure in all (or almost all) activities most of the day, nearly every day; may be subjective or observed by others

**Weight loss or gain**

Significant weight loss (without dieting) or gain (change of >5% body weight in a month), or decrease or increase in appetite nearly every day; in children, may be failure to gain weight as expected

**Insomnia or hypersomnia**

Nearly every day

**Psychomotor agitation or retardation**

Nearly every day and observable by others (not merely subjectively restless or slow)

**Fatigue**

Or loss of energy, nearly every day

**Feeling worthless or excessive/inappropriate guilt**

Nearly every day; guilt may be delusional; not merely self reproach or guilt about being sick

**Decreased concentration**

Nearly every day; may be indecisiveness; may be subjective or observed by others

**Thoughts of death/suicide**

Recurrent thoughts of death (not just fear of dying), recurrent suicidal ideation without specific plan, or suicide attempt, or a specific plan for suicide

Does grief qualify for major depression?

Stand up for **yes**

Sit down for **no**

Is major depression adaptive?

Stand up for **yes**

Sit down for **no**

Give it to me straight, doc

1. There is no objective boundary between 'normal' and 'abnormal'.
2. To explain something, you need to know if it qualifies as a trait that selection can operate on.



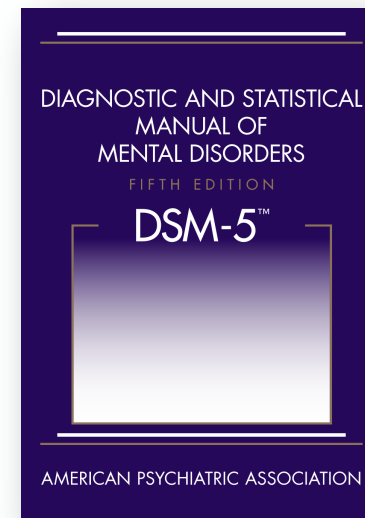
Biology is the study of continua + boundaries on behavioral continua are normal/abnormal
Where to draw the line?

What are we explaining from an evolutionary perspective?

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Why do we care?

WHO: Depression is the **leading cause of ill health and disability** worldwide.

Depression contributes more to DALYs (one 'lost' year of healthy life in the population) than almost all other sources of ill health

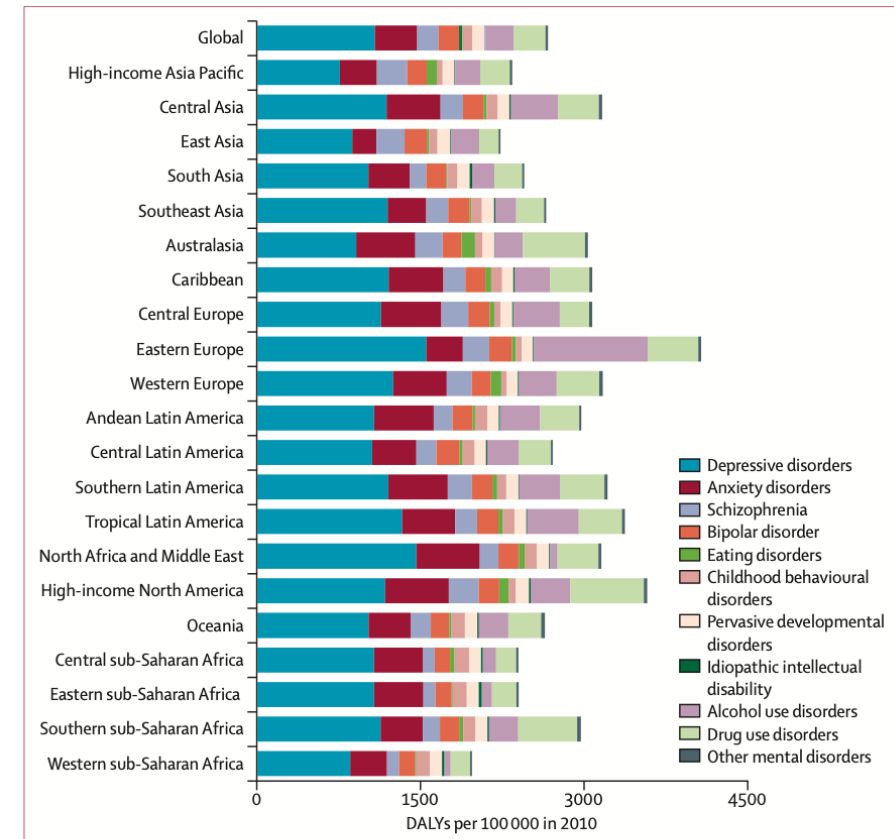


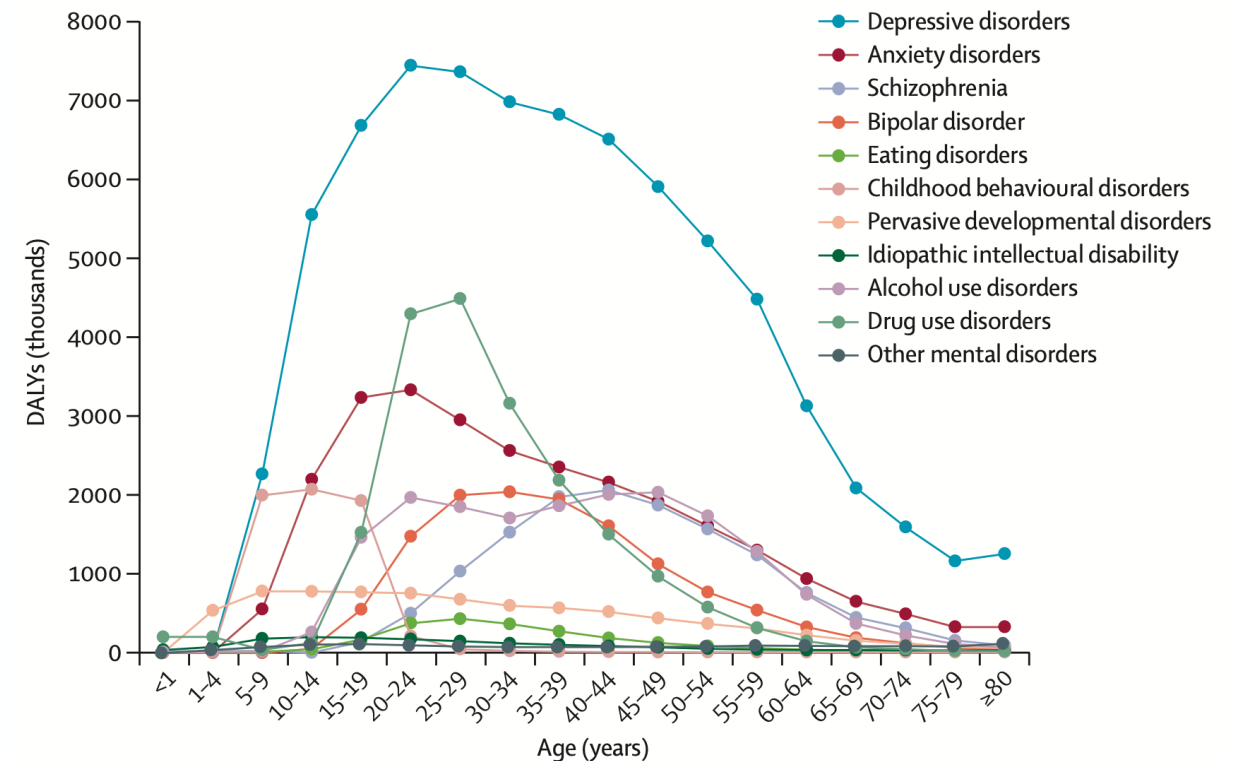
Figure 4: Rates of disability-adjusted life years (DALYs) per 100 000 individuals for mental and substance use disorders in 2010, by region

Why do we care?

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Depression contributes more to DALYs (one 'lost' year of healthy life in the population) than almost all other sources of ill health

From an evolutionary perspective: surprisingly early onset *and* therefore likely effect on fitness



Evolutionary explanations for major depression

Adaptive

Major depression is a good thing for fitness

Other animals (vertebrates & invertebrates) show depressive symptoms when mounting immune responses. In humans, we have the same but we experience subjective negative emotions.

Sickness behaviour

Depressive behaviours as million year old adaptive response

Raison, C. L., & Miller, A. H. (2013). *Molecular psychiatry*, 18(1), 15-37.

Managing risk-taking in adverse ecologies

Depression is adaptive b/c it changes how willing you are to take risks when is very costly in add'l envs

Nesse, R. M. (2013). *Depression*, 182-188.

Rumination and problem-solving

Depression is fixation on problems. Allows you to see all angles which could help you solve it.

Andrews, P. W., & Thomson Jr, J. A. (2009). *Psychological review*, 116(3), 620.

Disengagement from unattainable goals

Auto-correlated environment. People learning from persistent patterns in life → attainable/unattainable classification.

Nesse, R. M. (2013). *Depression*, 182-188.

Signalling and care-eliciting

Link to post-partum depression

Signaling to others that you need help to support

Hagen, E. H. (2003). *Genetic and cultural evolution of cooperation*, 95-123.

Depression is a signal to others to communicate status. Show to others that you are unable to cope & need help.

Energy conservation

But should it just be conceived via illness? What if conditions are bad so you need to conserve energy.

Watson, P. J., & Andrews, P. W. (2002). *Journal of affective disorders*, 72(1), 1-14.

Throwback Thursday

a	Proximate	Ultimate
Contemporary	Causation	Current Utility
Historical	Ontogeny	Evolution

Evolutionary explanations for major depression

Adaptive

**Sickness
behaviour**

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**Managing
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**Signalling
and care-
eliciting**

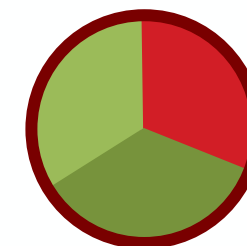
Hagen, E. H. (2003). *Genetic and cultural evolution of cooperation*, 95-123.

**Energy
conservation**

Watson, P. J., & Andrews, P. W. (2002). *Journal of affective disorders*, 72(1), 1-14.

Throwback Thursday

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Finite energy

These allow you to conserve energy, depression as adaptive means to balance allocation b/wn maint + and reproduction

Evolutionary explanations for major depression

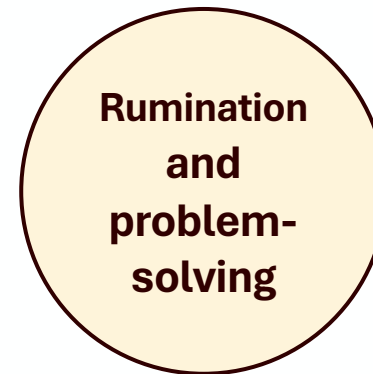
Adaptive



Raison, C. L., & Miller, A. H. (2013). *Molecular psychiatry*, 18(1), 15-37.



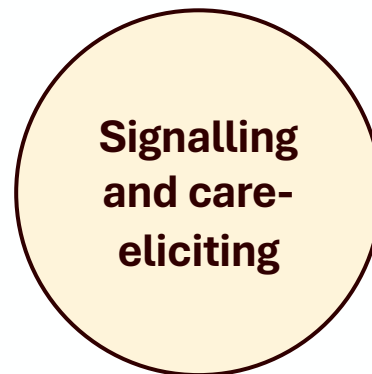
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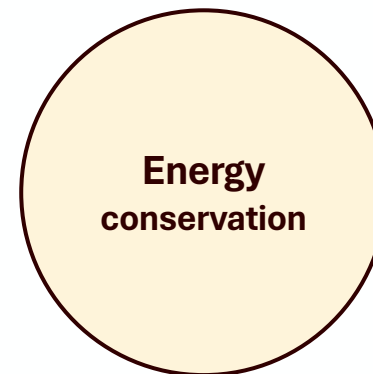
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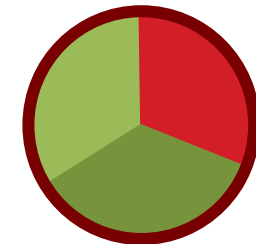
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Throwback Thursday

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Finite energy

poor health - would make sense to be depressed

Context is key

Important clinical and public health implications

Problems with evolutionary explanations for major depression

1. Multiplicity

(there are many – which one is correct?) → which explains most cases?

2. All assume major depression is adaptive

3. We've assumed 'major depression' is a *real* 'thing' that evolution acted on

Rumination
and
problem-
solving

Irews, P. W., & Thomson Jr, J. A.
Psychological review, 116(3), 620.

IS de pression
actually a trait
natural selection
can act on?

Throwback Thursday

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Neuroscience News

Genetic Roots of Depression Reveal Strong Suicide Risk Signals

Depression in young adulthood has a stronger hereditary component and is associated with a higher risk of suicide attempts than depression that...

3 weeks ago

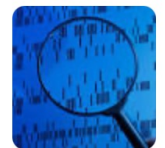


The Guardian

Women carry a higher genetic risk of depression, new study says

Women carry a higher genetic risk of depression, new study says ... Women carry a higher genetic risk of depression, a new study has found.

Oct 7, 2025

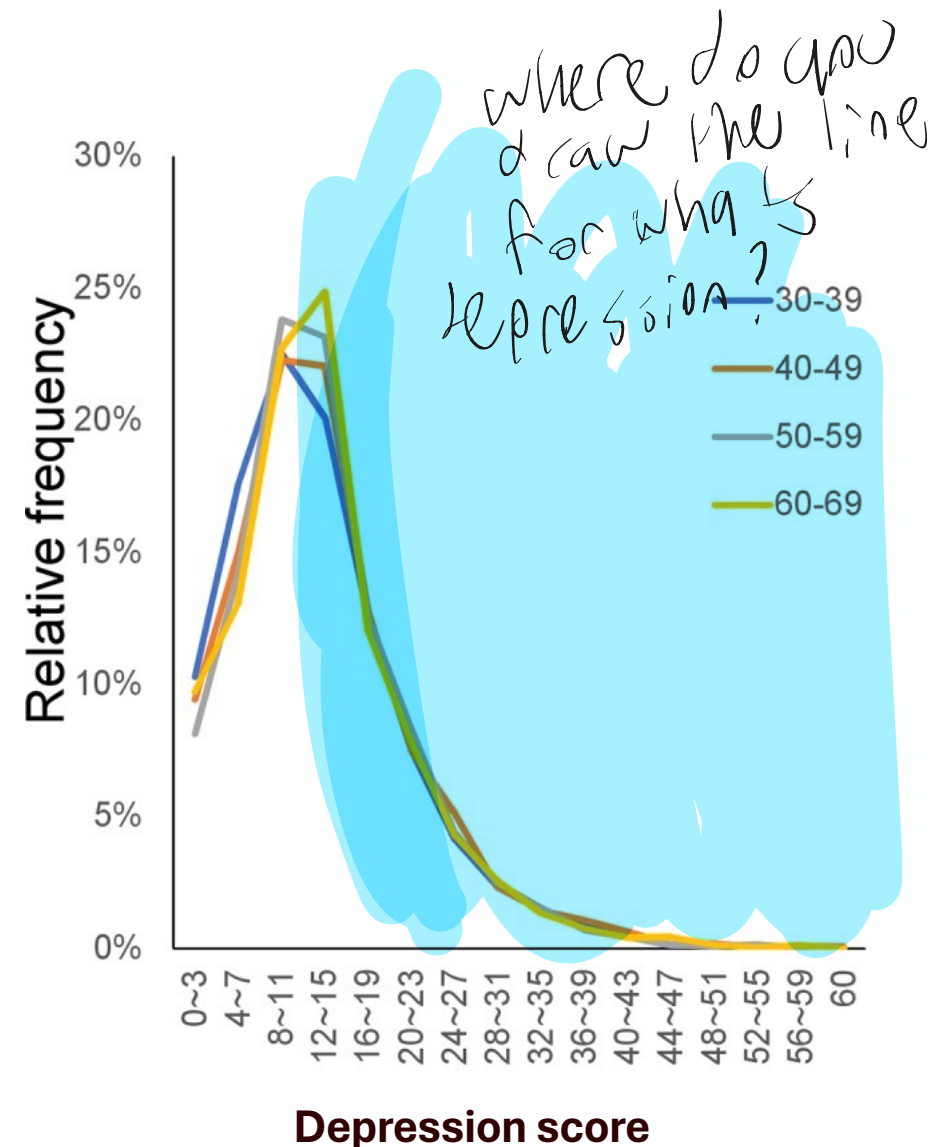


What *is* major depression?

Depressive symptoms

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Recurrent thoughts of death (not just fear of dying), recurrent suicidal ideation without specific plan, or suicide attempt, or a specific plan for suicide



Key

Major depression is a 'latent variable'

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Group of symptoms
we've given a name
also have for GAD

Decreased concentration

Psychomotor agitation

Depressed mood

'Major depression'

Anhedonia

Hypersomnia

Weight gain

Both these sets of symptoms would warrant MDD dx

Excessive guilt

Psychomotor retardation

Depressed mood

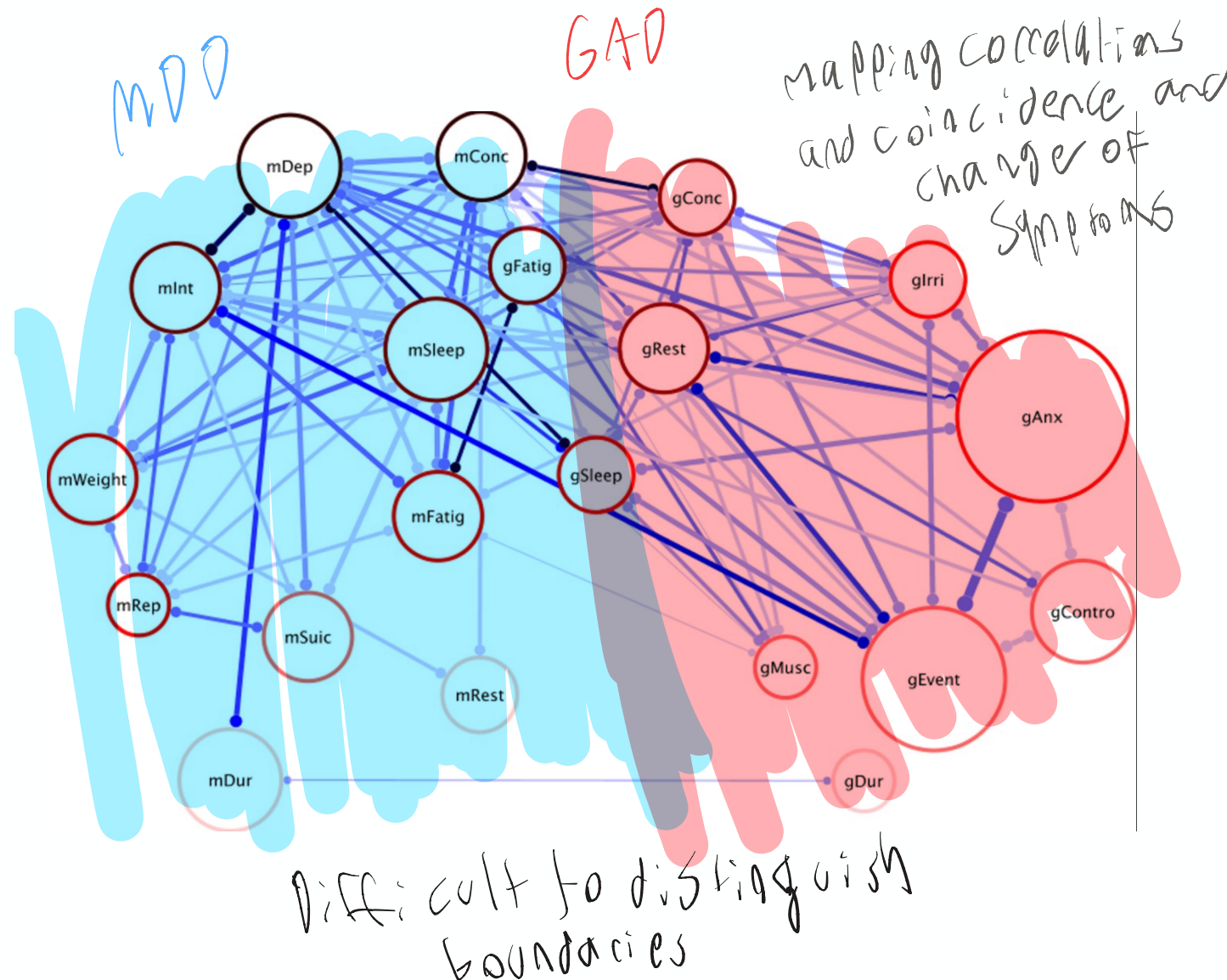
'Major depression'

Anhedonia

Insomnia

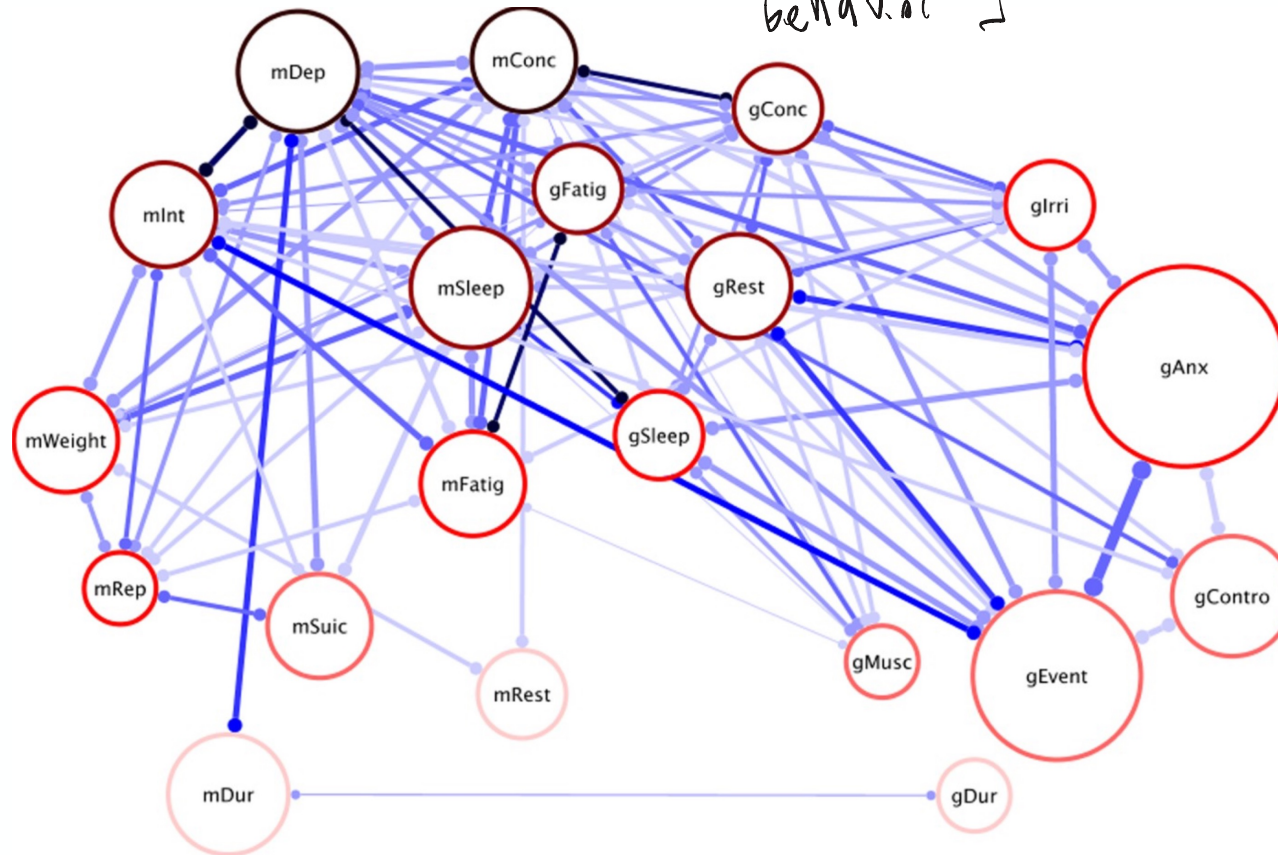
Weight loss

Symptomics – a burgeoning area of research and treatment



Thinking in terms of mechanisms

mechanisms to alter behavior } group of mechanisms



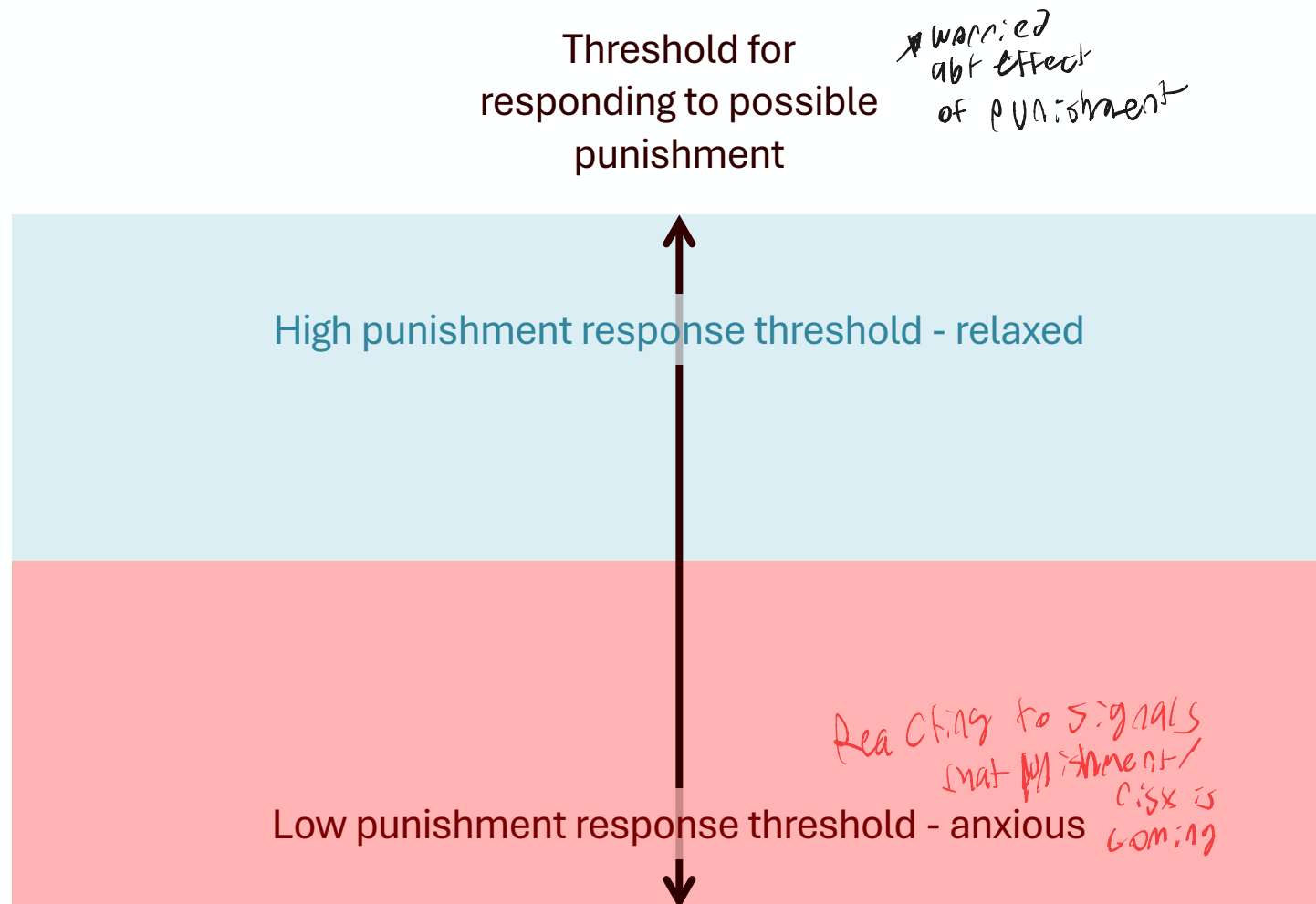
were
to think
in
proximal
terms

Throwback Thursday

a

	Proximate	Ultimate
Contemporary	Causation	Current Utility
Historical	Ontogeny	Evolution

These mechanisms are there to get you to do (or *not do*) something

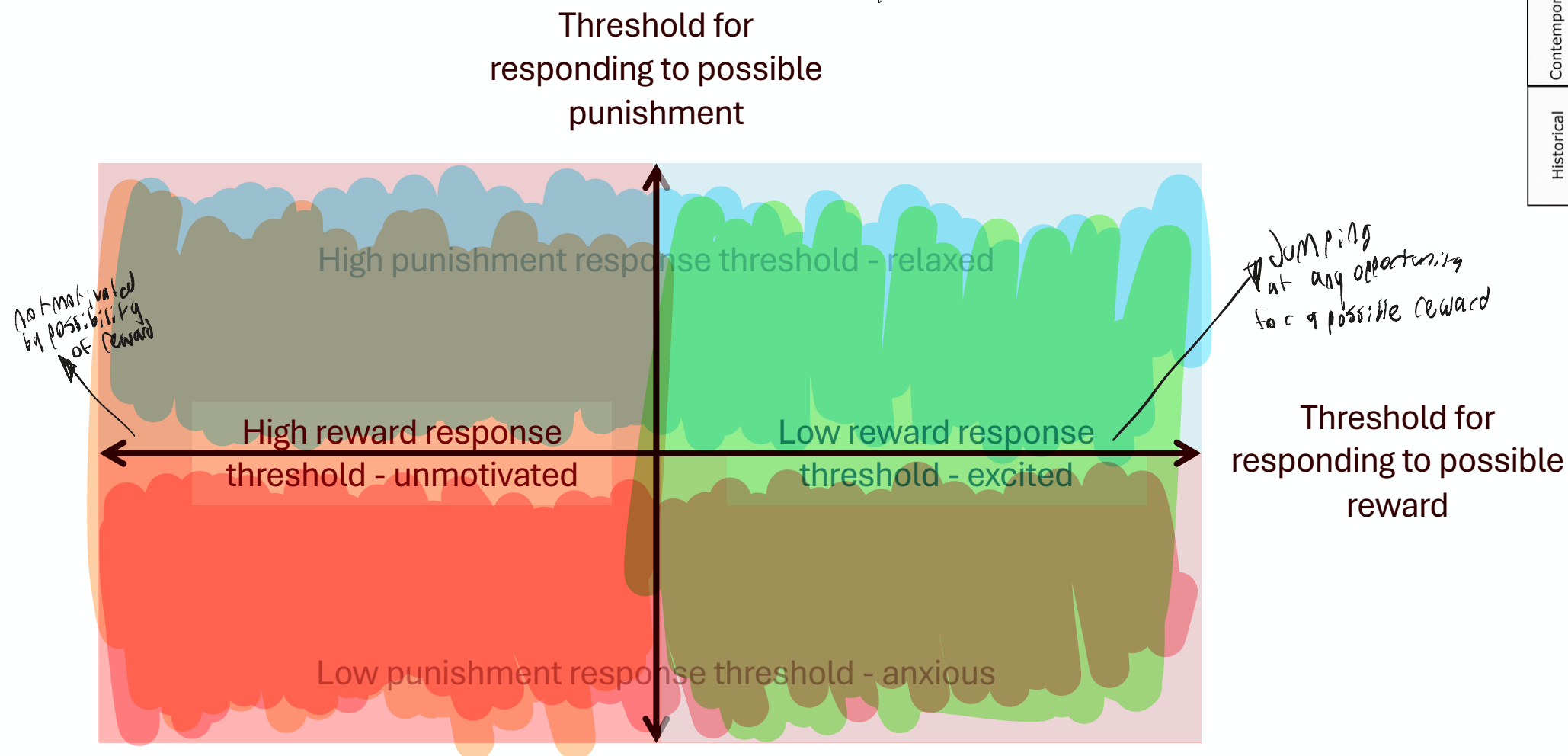


Throwback Thursday

a	Proximate	Ultimate
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	Calculation <i>Caution</i>	Current Utility
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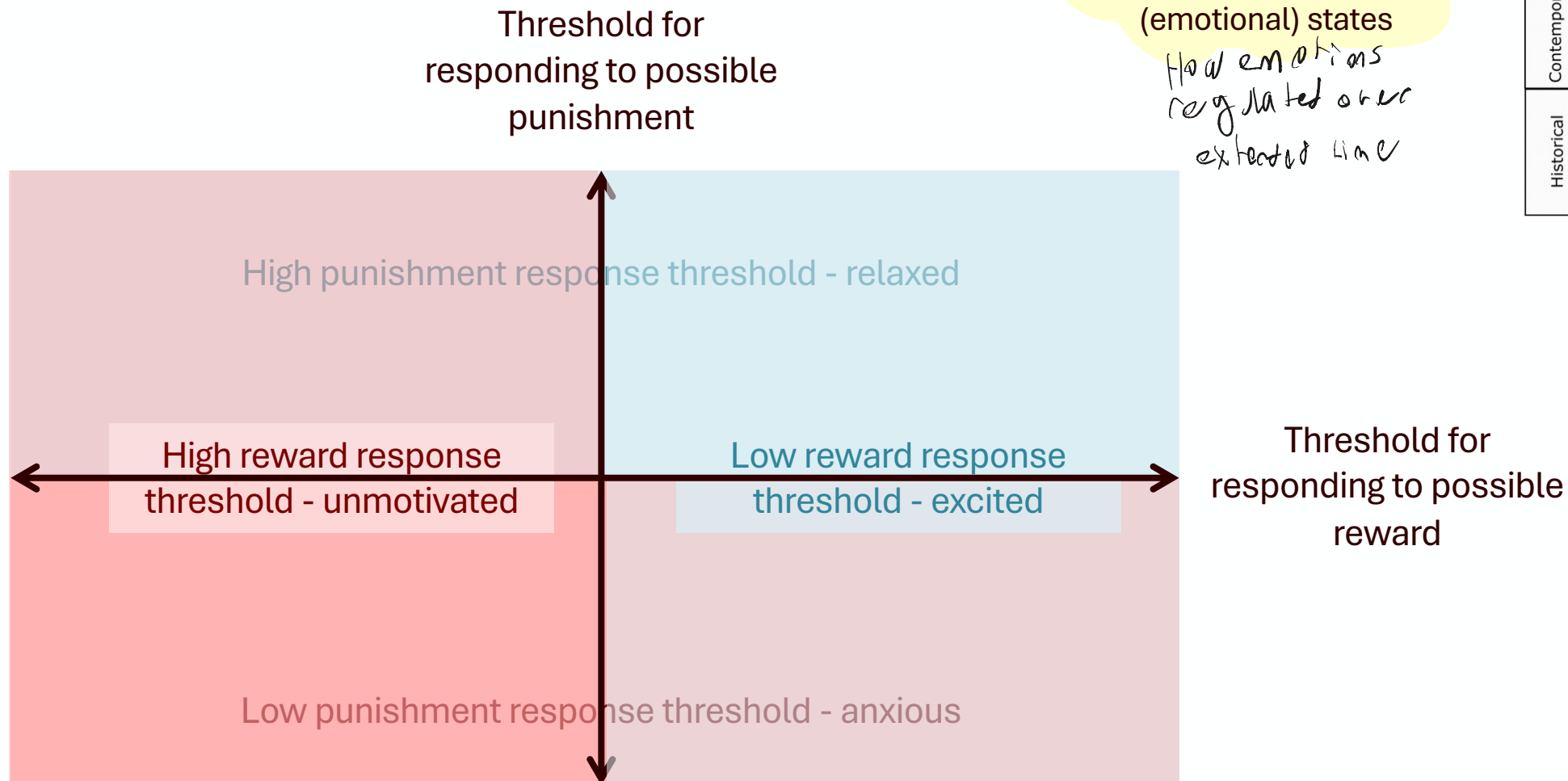
Matrix
For classifying mood



Throwback Thursday

a	Proximate	Ultimate
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Historical	Ontogeny	Evolution

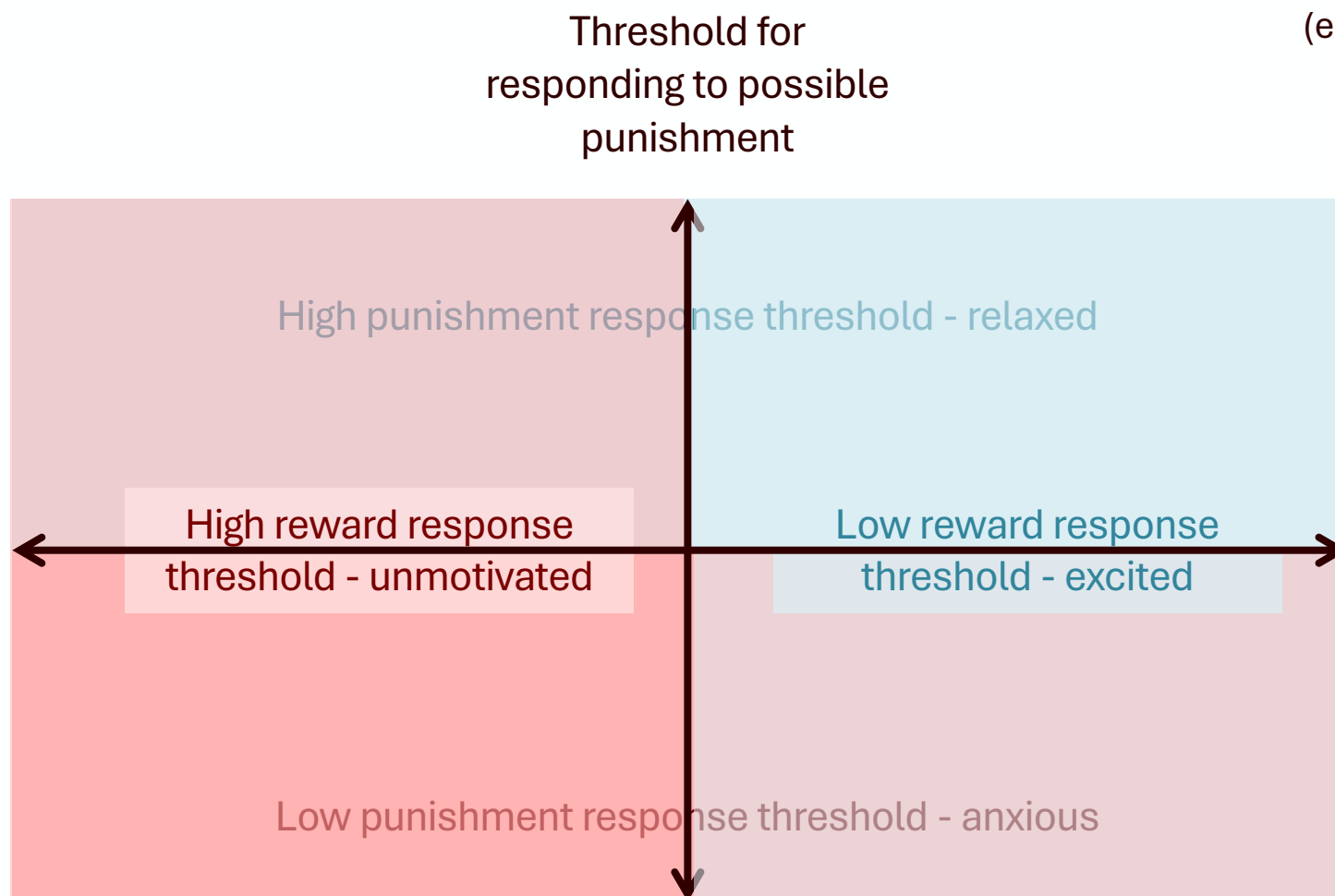
Biaxial framework for classifying mood



Throwback Thursday

a	Proximate	Ultimate
	Calculation	Current Utility
Contemporary		
Historical	Ontogeny	Evolution

Biaxial framework for classifying mood

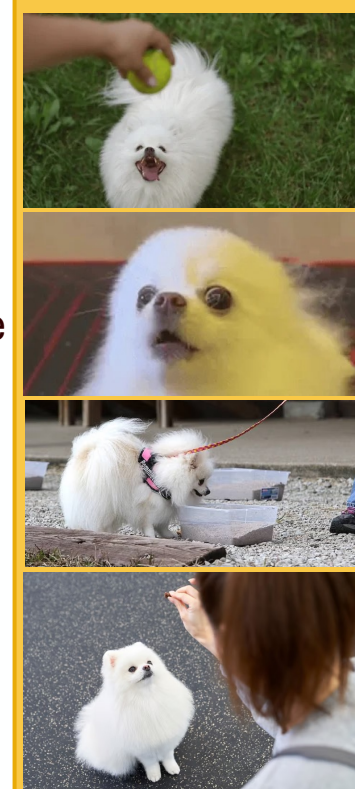
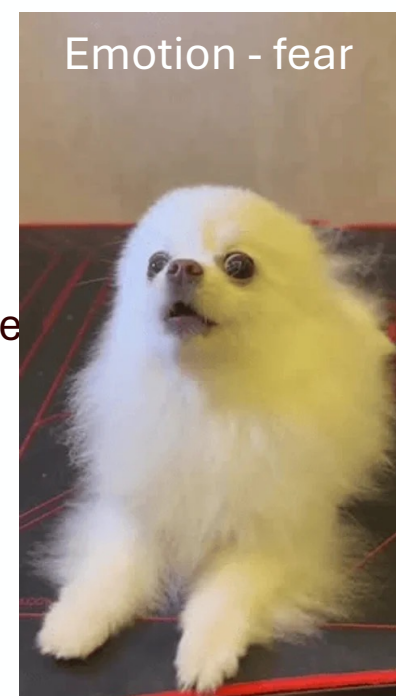


Enduring affective (emotional) states

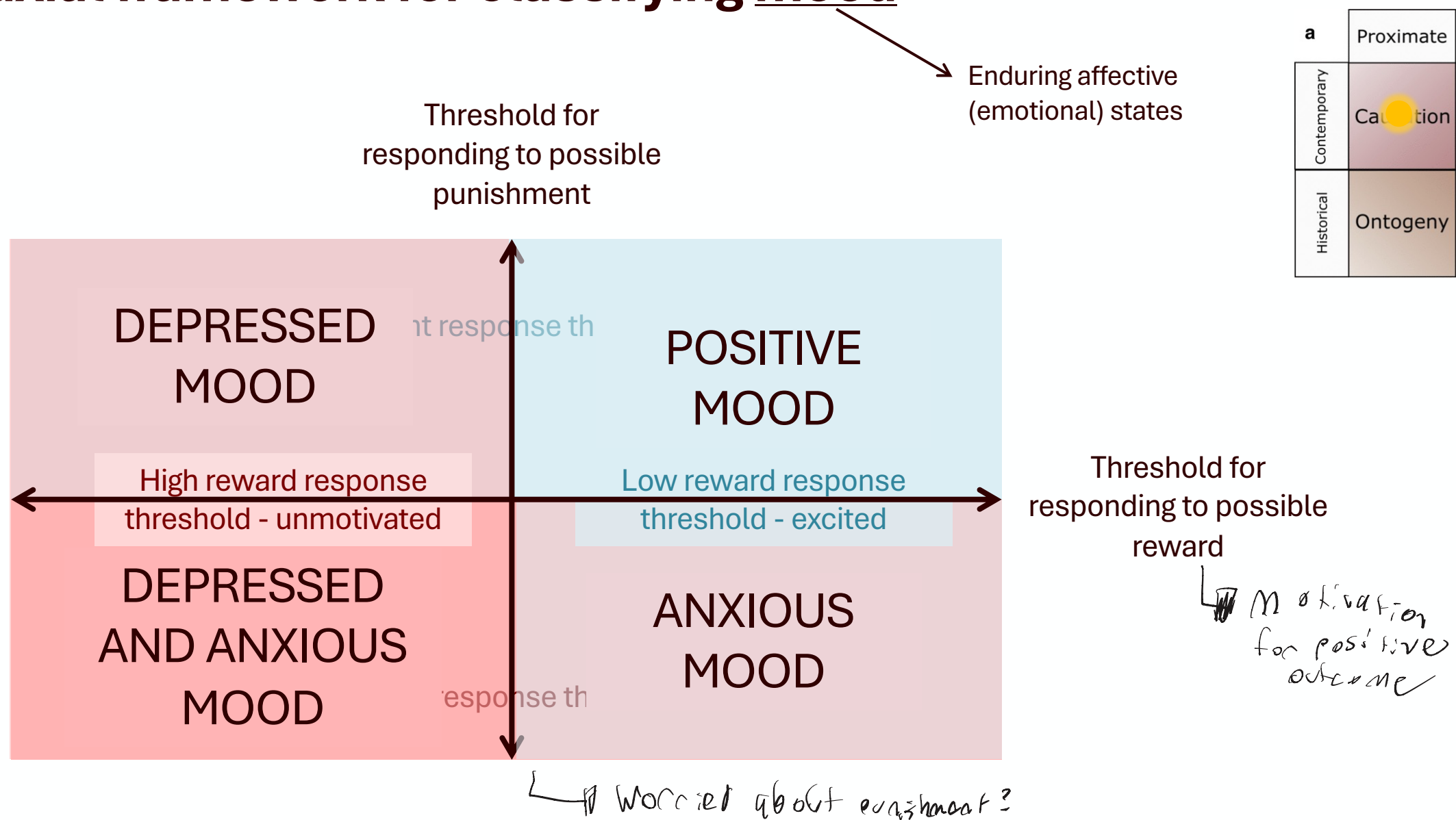
Not emotions
but persisting
states

Throwback Thursday

a	Proximate	Ultimate
	Calculation	Current Utility
Contemporary		
Historical		



Biaxial framework for classifying mood



Throwback Thursday

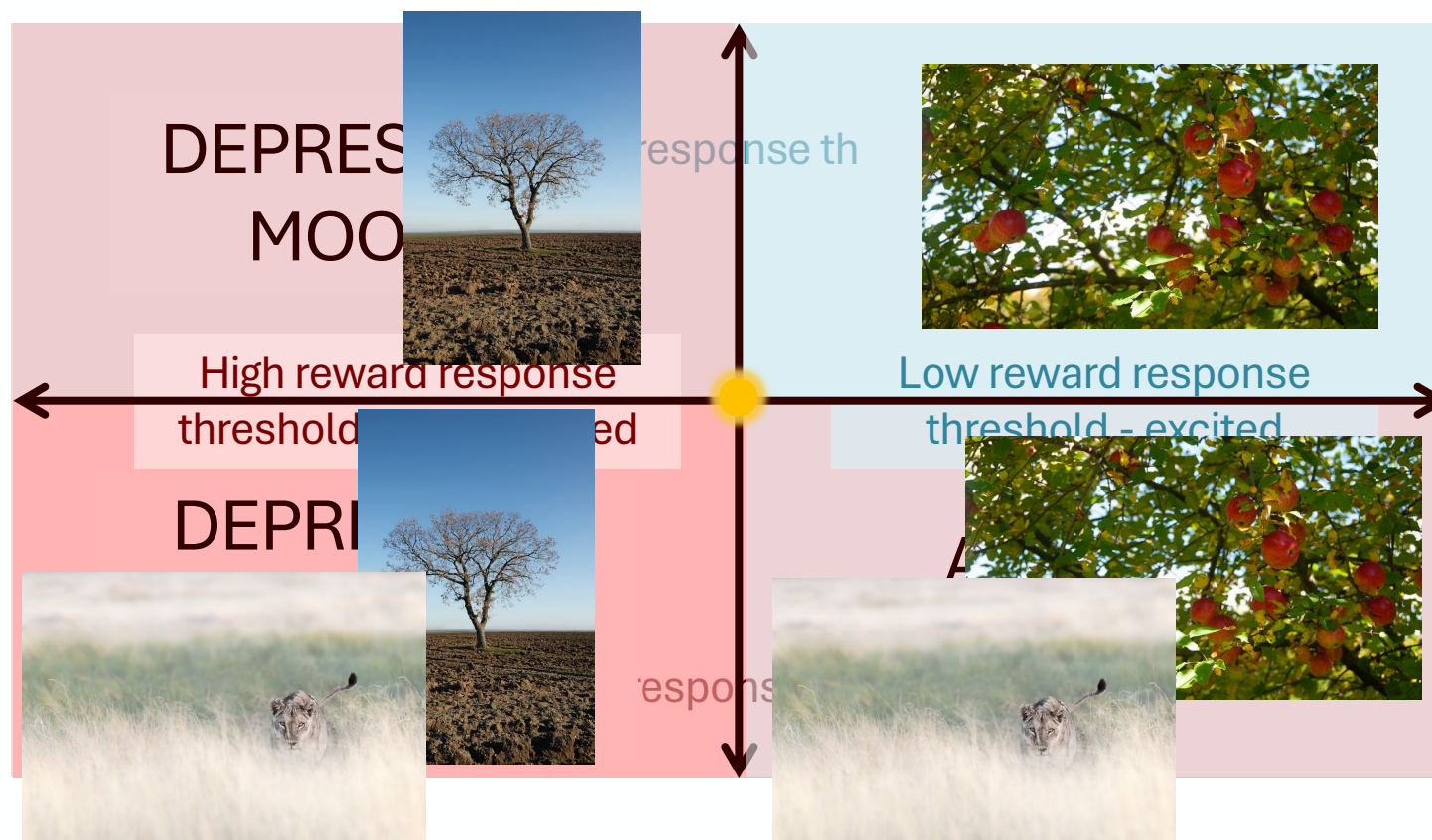
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Biaxial framework for classifying mood

Enduring affective
(emotional) states

Throwback Thursday

a	Proximate	Ultimate
	Contemporary Calibration	Current Utility
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- Over time, usually homeostatic
- **A form of nongenetic adaptation – calibrating behaviour to**
 - **The environment**
 - **The organism's status**

Not plasticity b/c
not physical.
Behavioural adaptation
to calibrate avg
behavior to individual
health and environment
possibilities
of punishment
or reward
Impact of
punishment
and ability
to obtain
reward

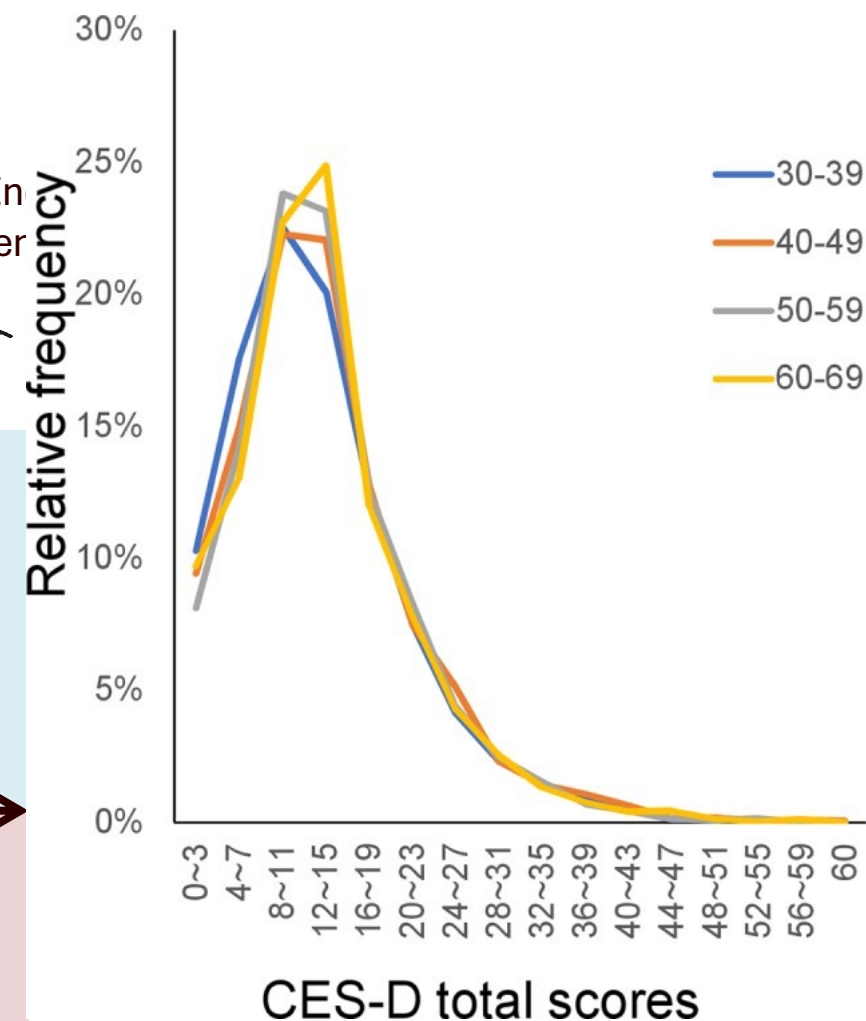
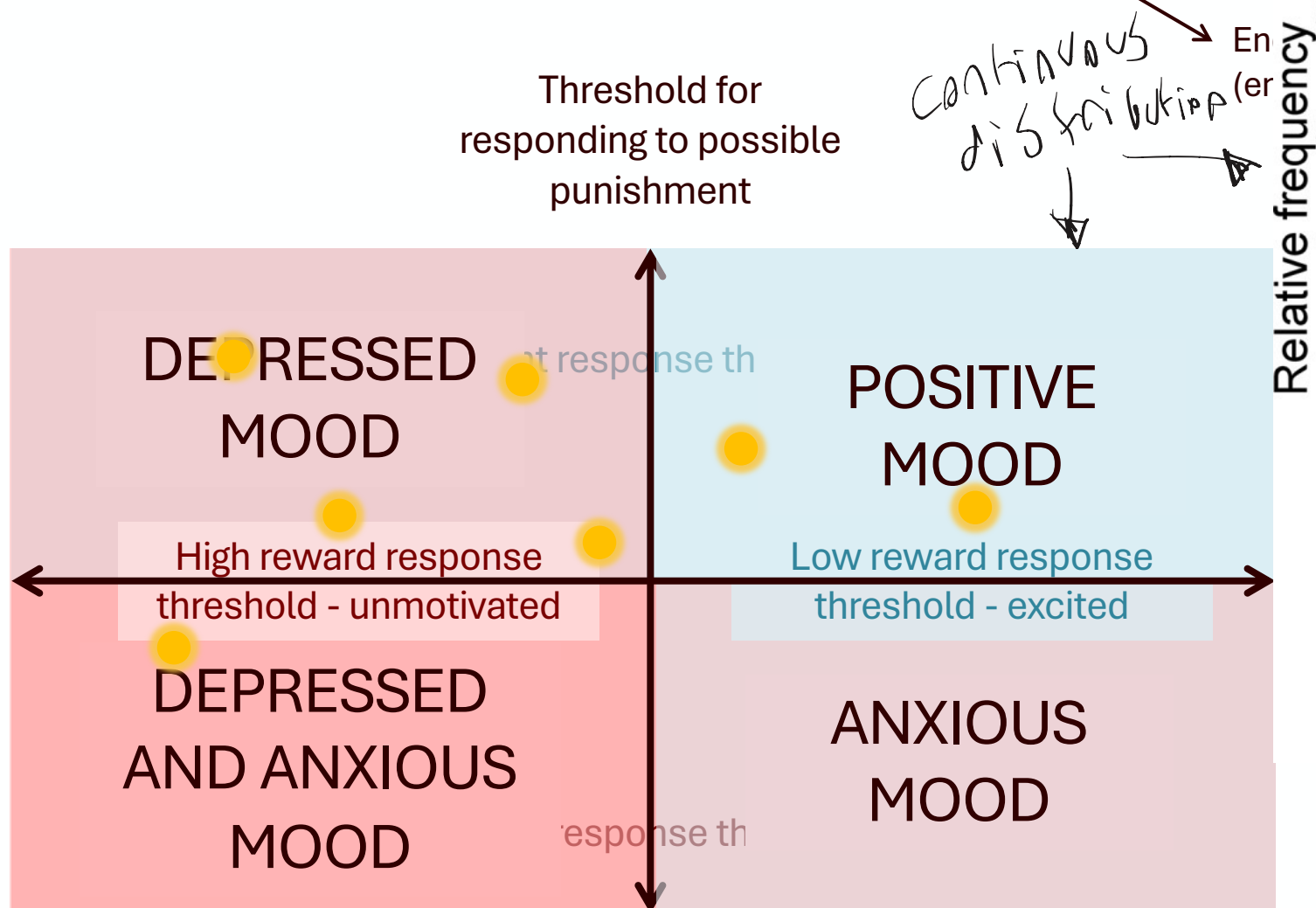
Can a chimpanzee get depressed?

WRT mood system
Lamer work, yes (not based on
subjective experience)
↳ you can have depressive
behaviors in this Lamer work!

Can a bacterium get depressed?

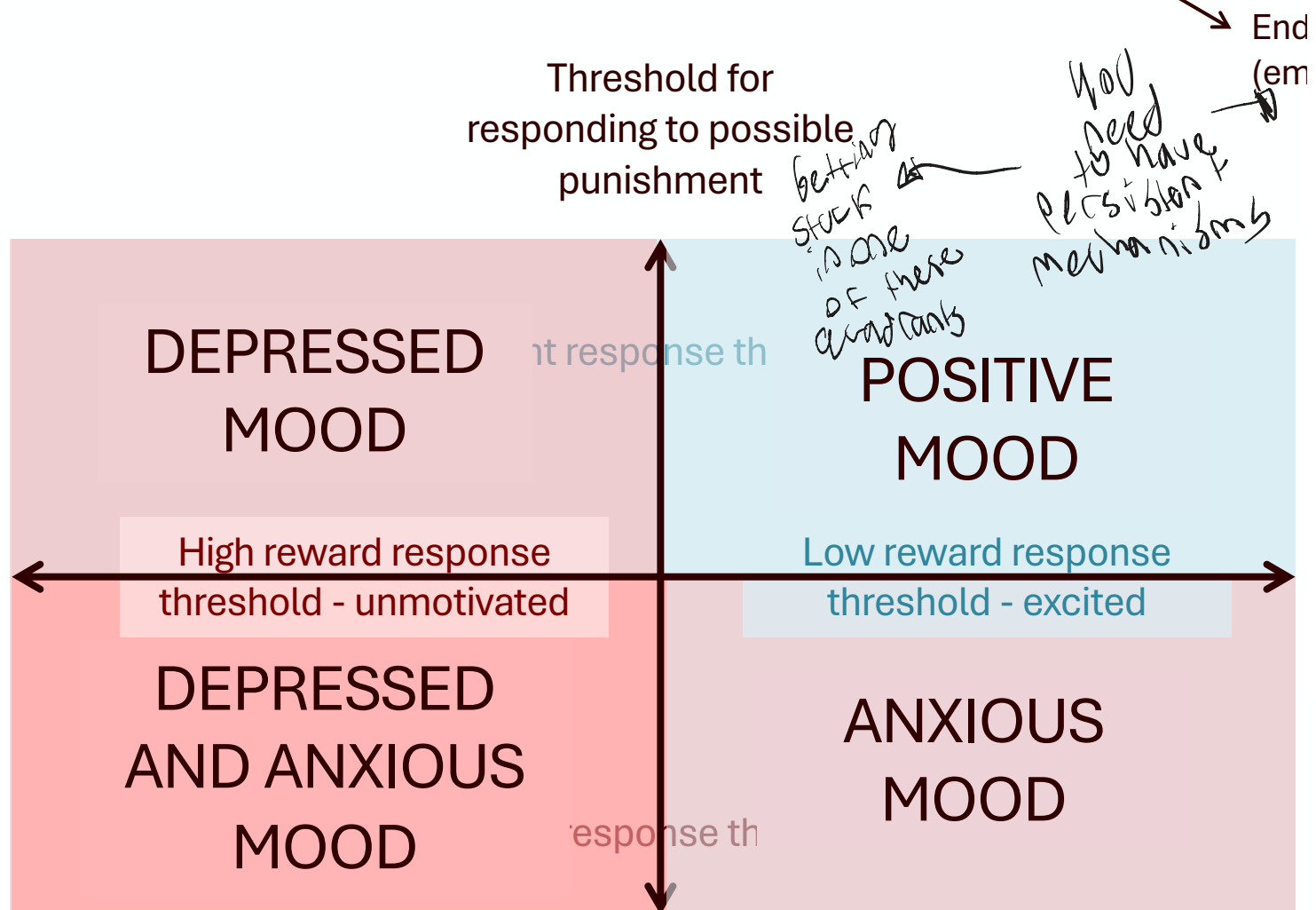
IN Mood Framework
Yes - depending on how
Yes or No - response time

Biaxial framework for classifying mood



Tomitaka, Shinichiro, et al. *Frontiers in psychiatry* 9 (2018): 390.

Biaxial framework for classifying mood



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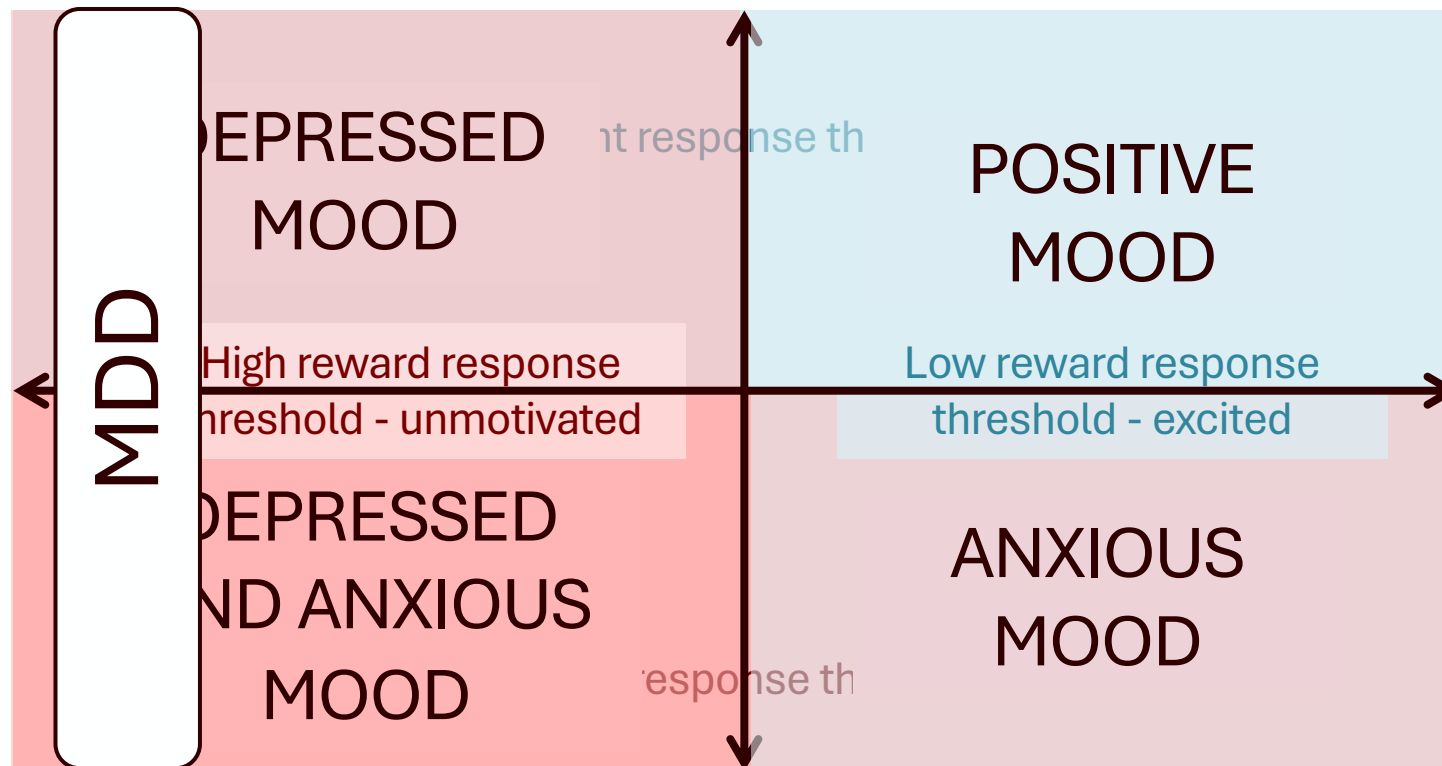
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Biaxial framework for classifying mood



Threshold for
responding to possible
punishment

Enduring affective
(emotional) states



Throwback Thursday

a	Proximate	Ultimate
	Contemporary Calculation	Current Utility
Historical	Ontogeny	Evolution

Key

Checking back in

- What we call 'major depression' is not, itself, a trait that was selected for
 - Hence no gene for depression!
- Rather: a *state* that the mood system can be in
 - Mood system comprises many mechanisms
 - Mood system is adaptive because it allows animals to calibrate their behaviour to their environment
 - Mood system is highly conserved
 - Genes can code for aspects of this system – e.g., sensitivity
- This state is not easily distinguishable from other states (positions in that diagram) – and the cutoff is a bit arbitrary



Mechanism
Example

Throwback Thursday

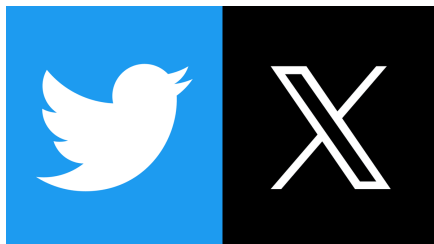
a		Proximate	Ultimate
		Contemporary	Historical
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'Depression'

Symptoms

a		Proximate	Ultimate
		Contemporary	Historical
	Contemporary	Causation	Current Utility
	Historical	Ontogeny	Evolution

'Hypersomnia'

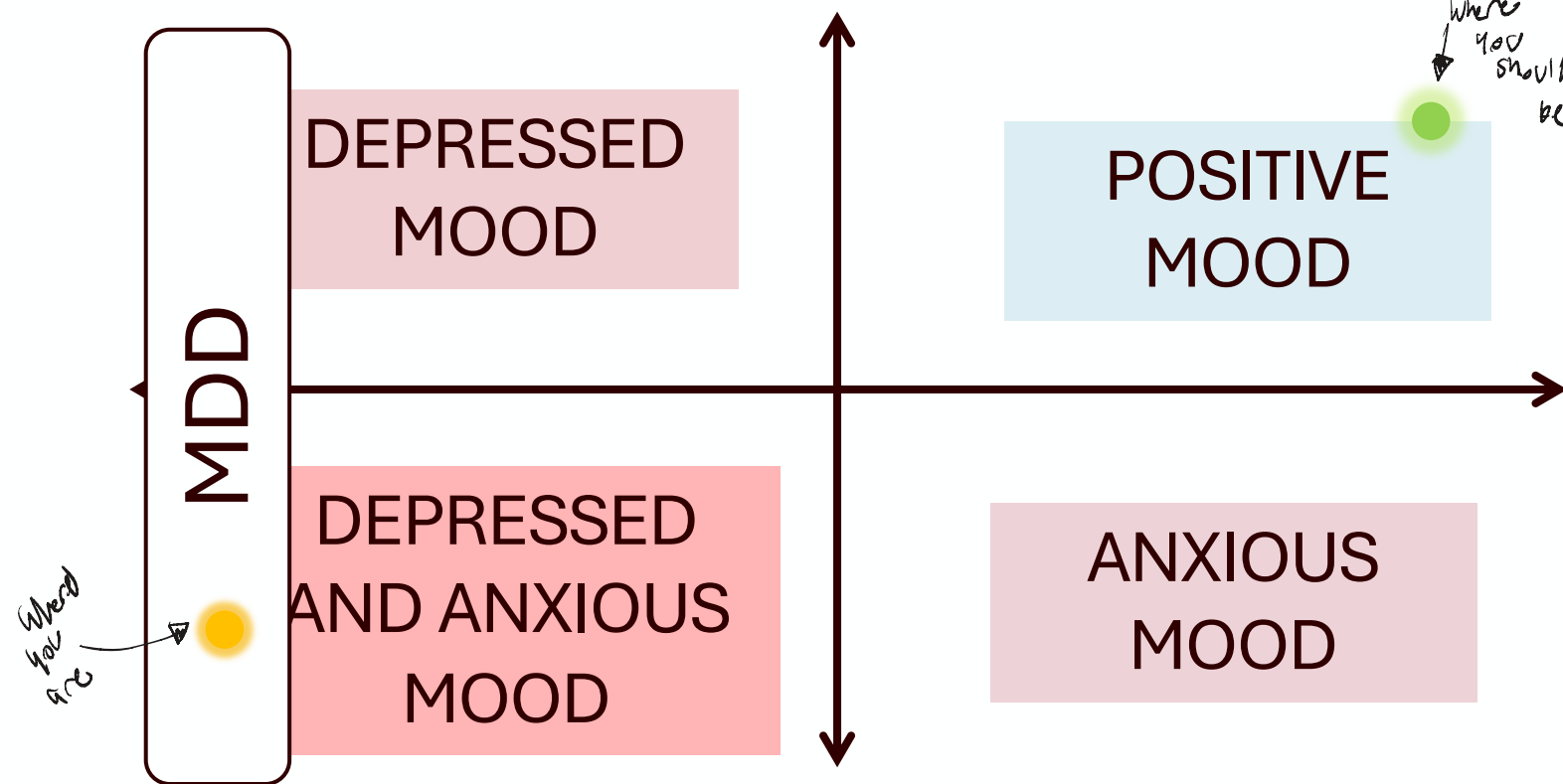
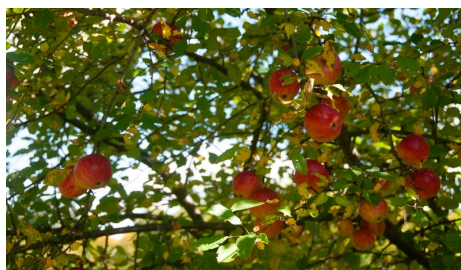


Throwback Thursday
Mismatch

Implications of this view

Why does depression happen?

1 Malfunctioning mood system



Throwback Thursday

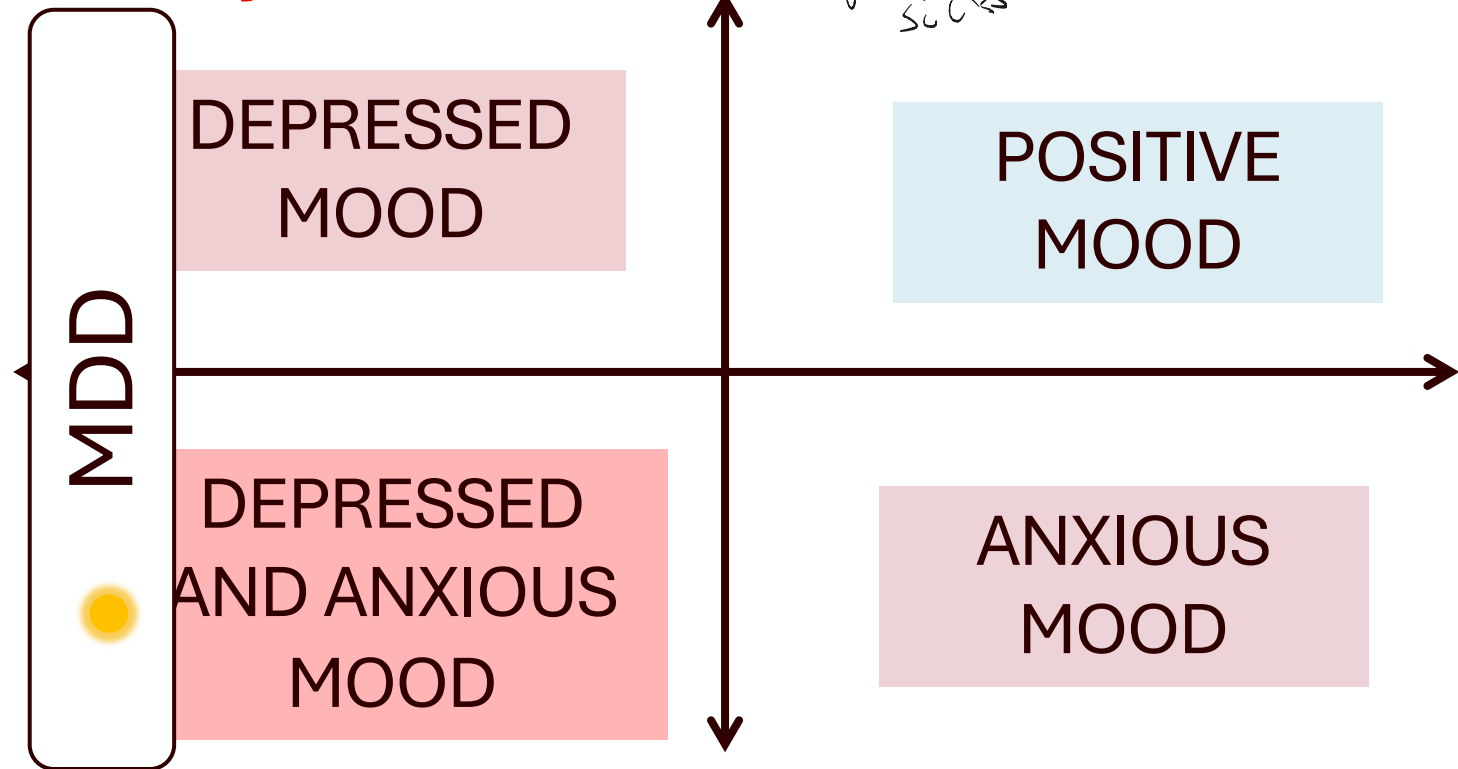
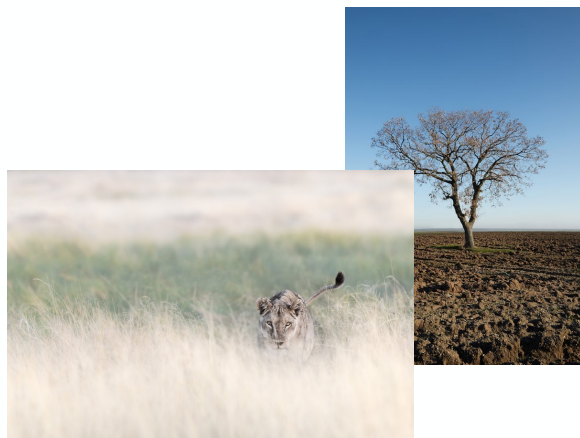
Mismatch

Implications of this view

Why does depression happen?

1 **Malfunctioning mood system**

2 **Correctly functioning mood system**



Throwback Thursday

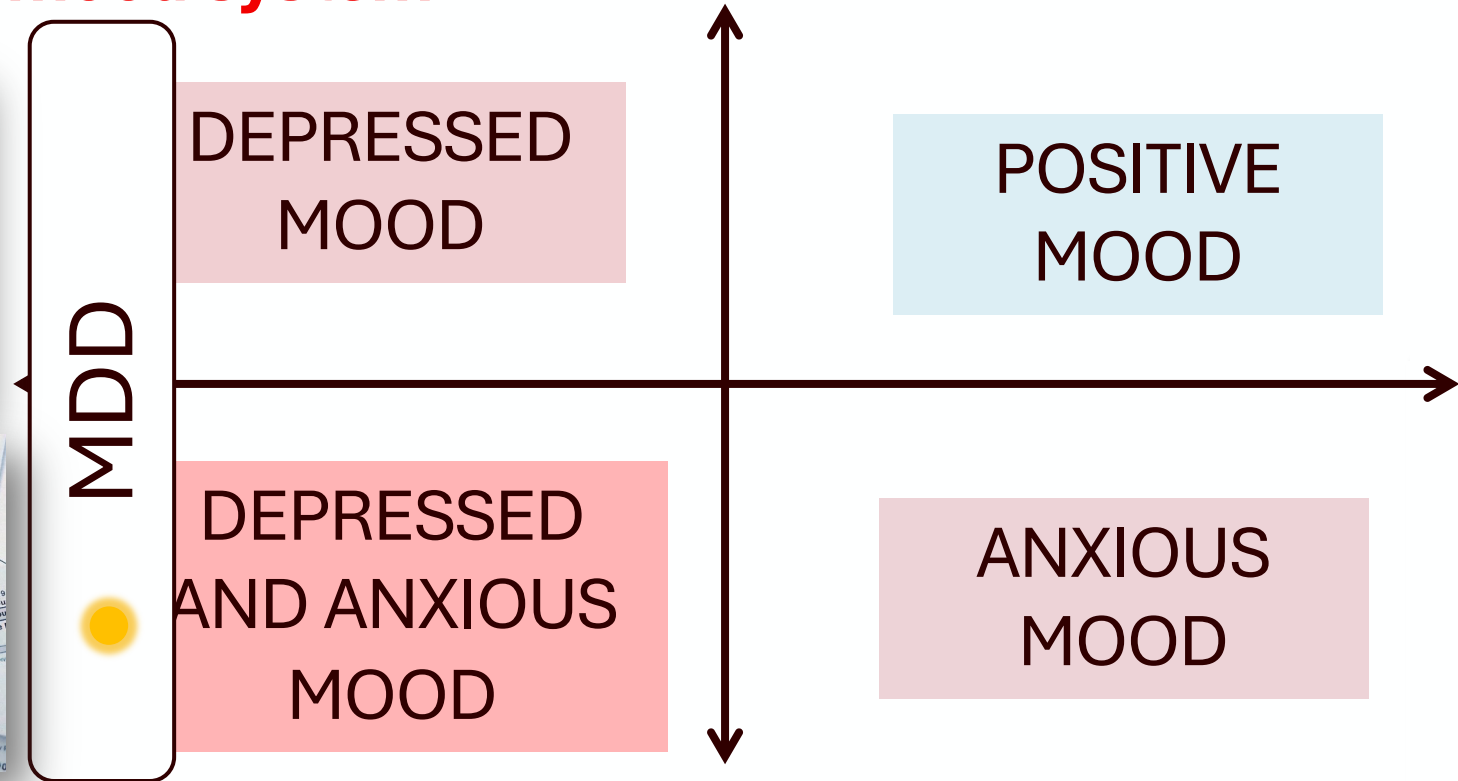
Mismatch

Implications of this view

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Throwback Thursday

Mismatch**Biosocial context**

Implications of this view

Why does depression happen?

1 **Malfunctioning mood system**

2 **Correctly functioning mood system**

Should we *not* treat the mechanisms that are grouped under depression, then?

Of course not. An evolutionary perspective is not the only one – suffering must be taken into account.

Evolutionary perspective: change the environment!

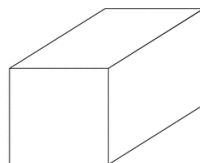
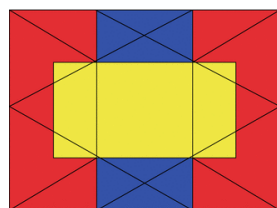
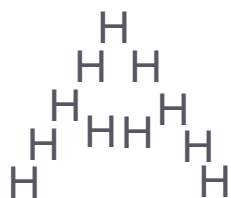


Can a chimpanzee get depressed?

Can a bacterium get depressed?



Autism, at its core, is difficulties in social and communicative development, alongside strong, narrow interests and repetitive behaviour.



Artistic
people tend
to identify
detail is
better

More connections
in brain & less
synaptic pruning

Continuum is
spectrum,
which ASD is

1. sarcastic

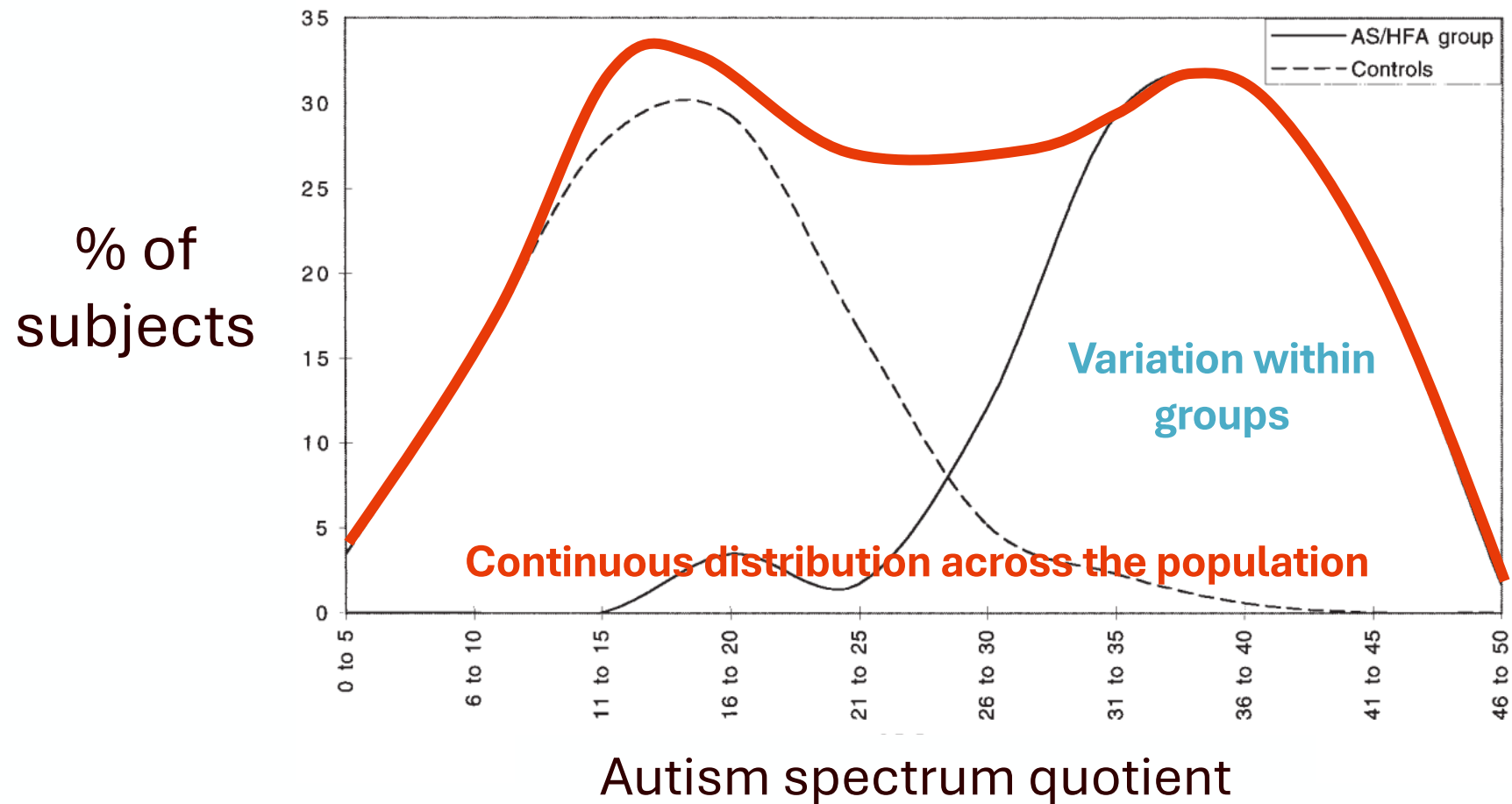
2. stern



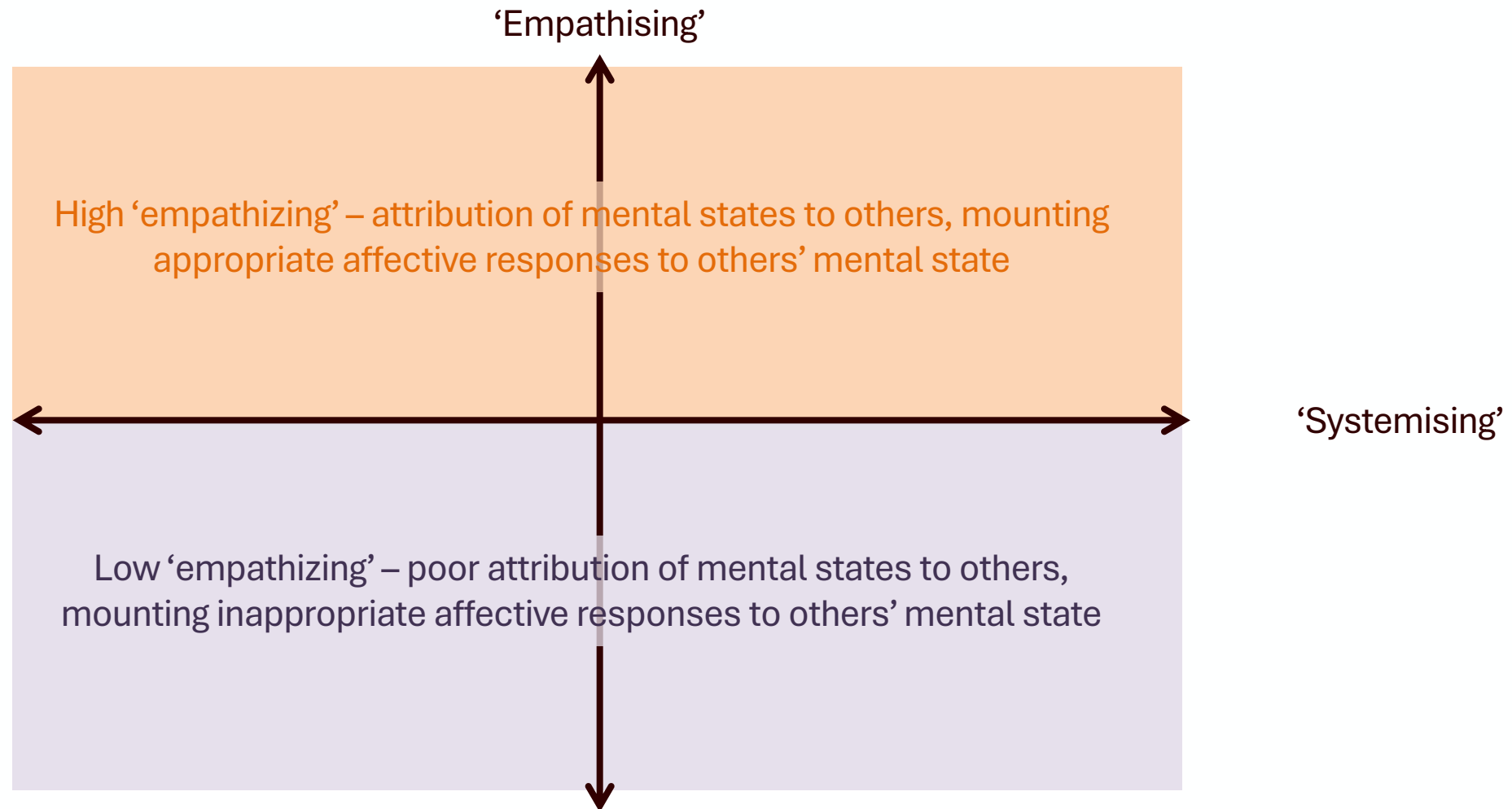
3. suspicious

4. dispirited

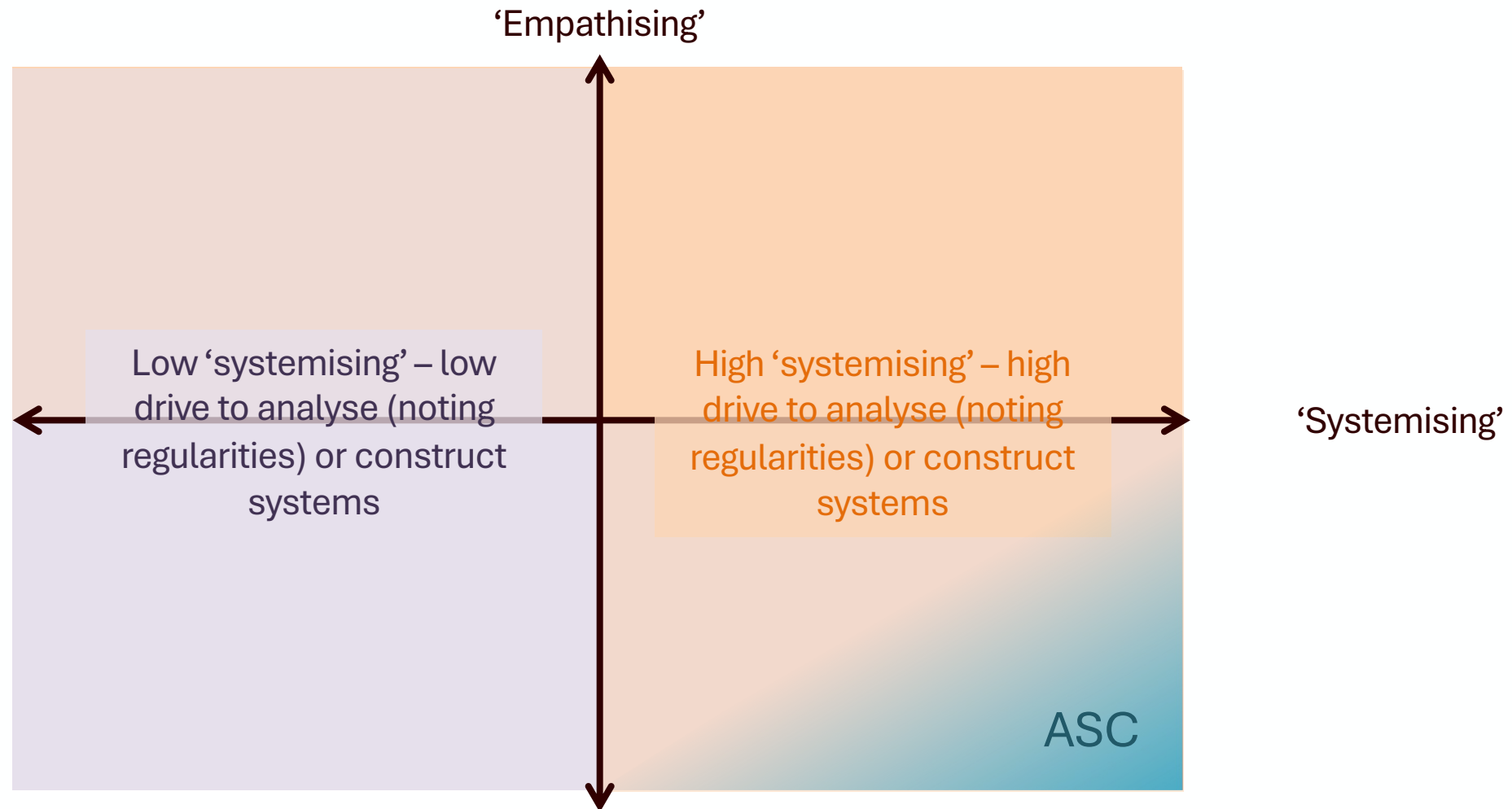
Scores of autism spectrum quotient illustrate the continuum



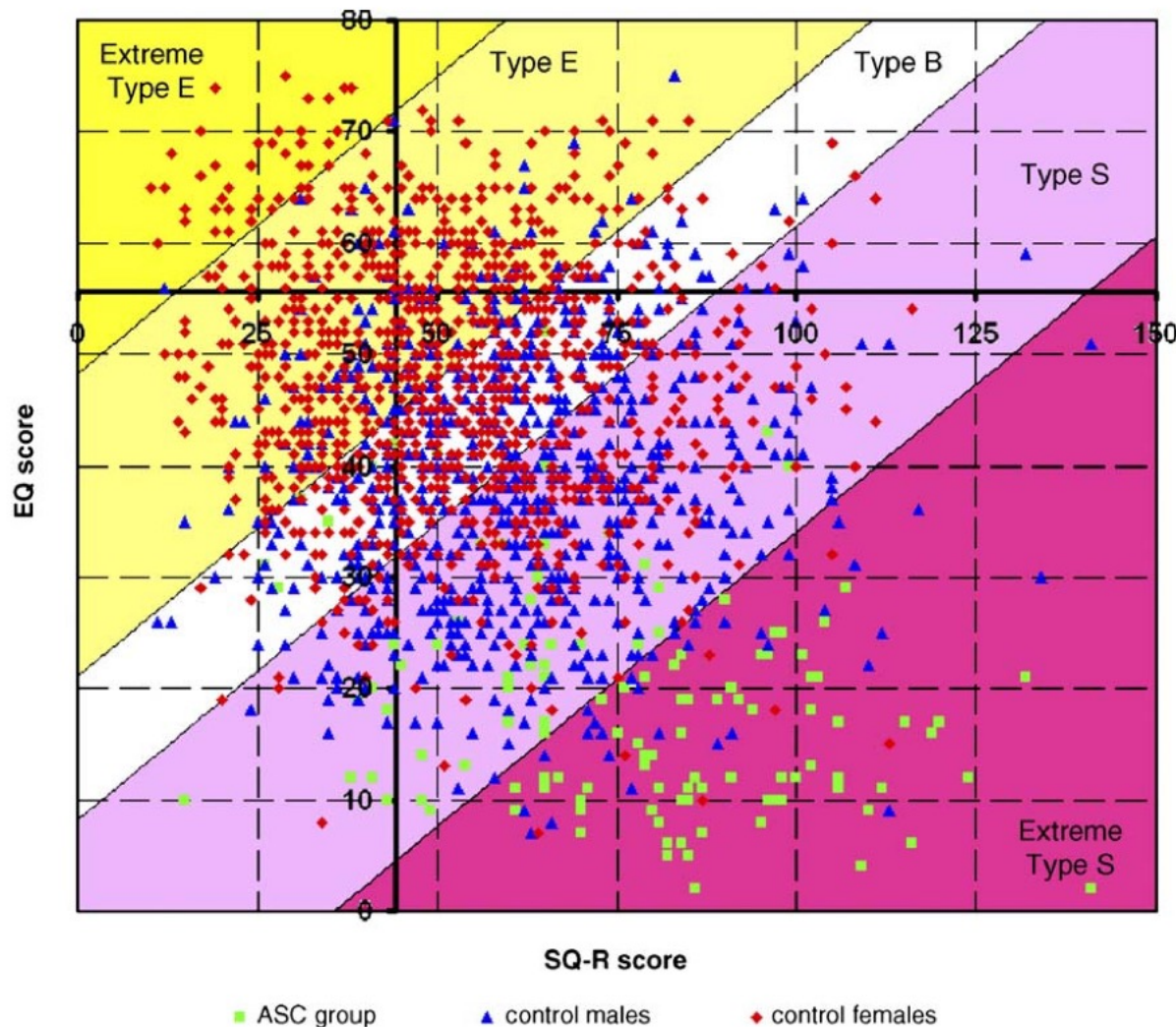
A biaxial framework for thinking about ASC



A biaxial framework for thinking about ASC



A biaxial framework for thinking about ASC: in practice



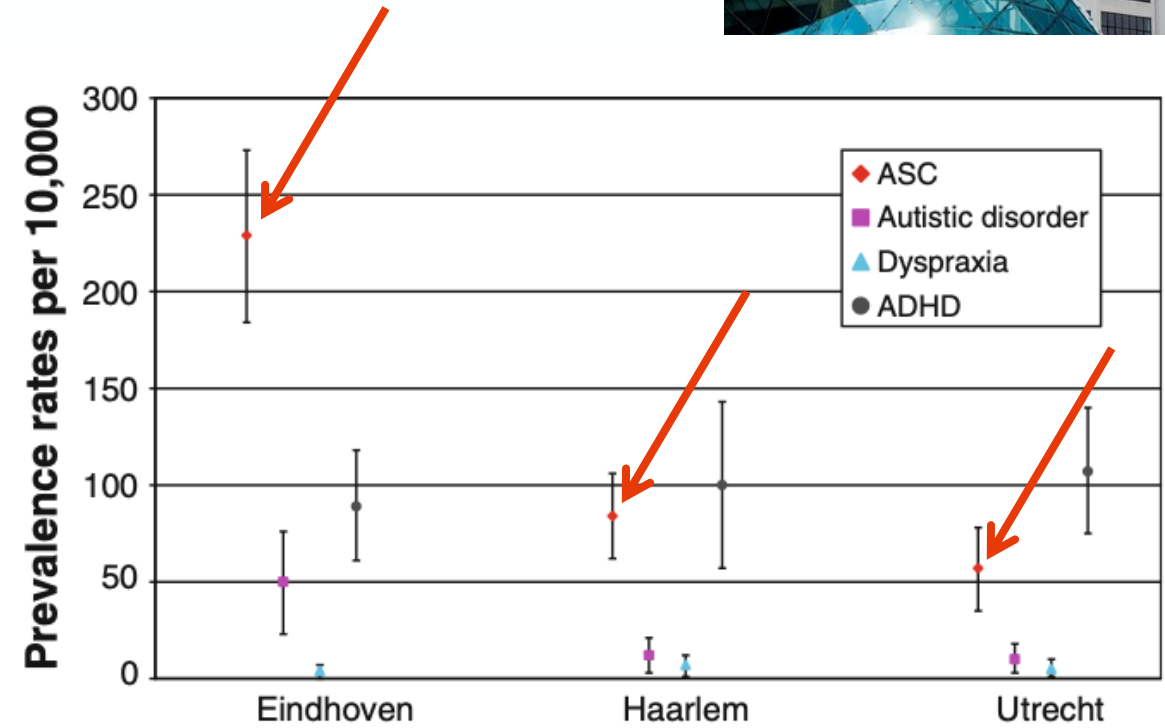
Learn path ising systematizing
axis positions
tend to be static

Important takeaways:

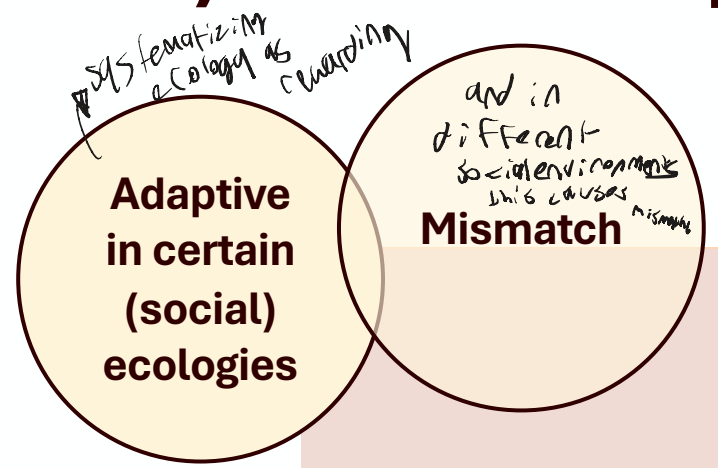
- Once again, a continuum!
 - Part of the normal distribution – not really a ‘disorder’ (unless it produces suffering).
- ASC is characterized by **high systematizing**
- In contrast to the mood framework, these points are (basically) static during life...

ASC is highly heritable: estimates are usually around 0.8

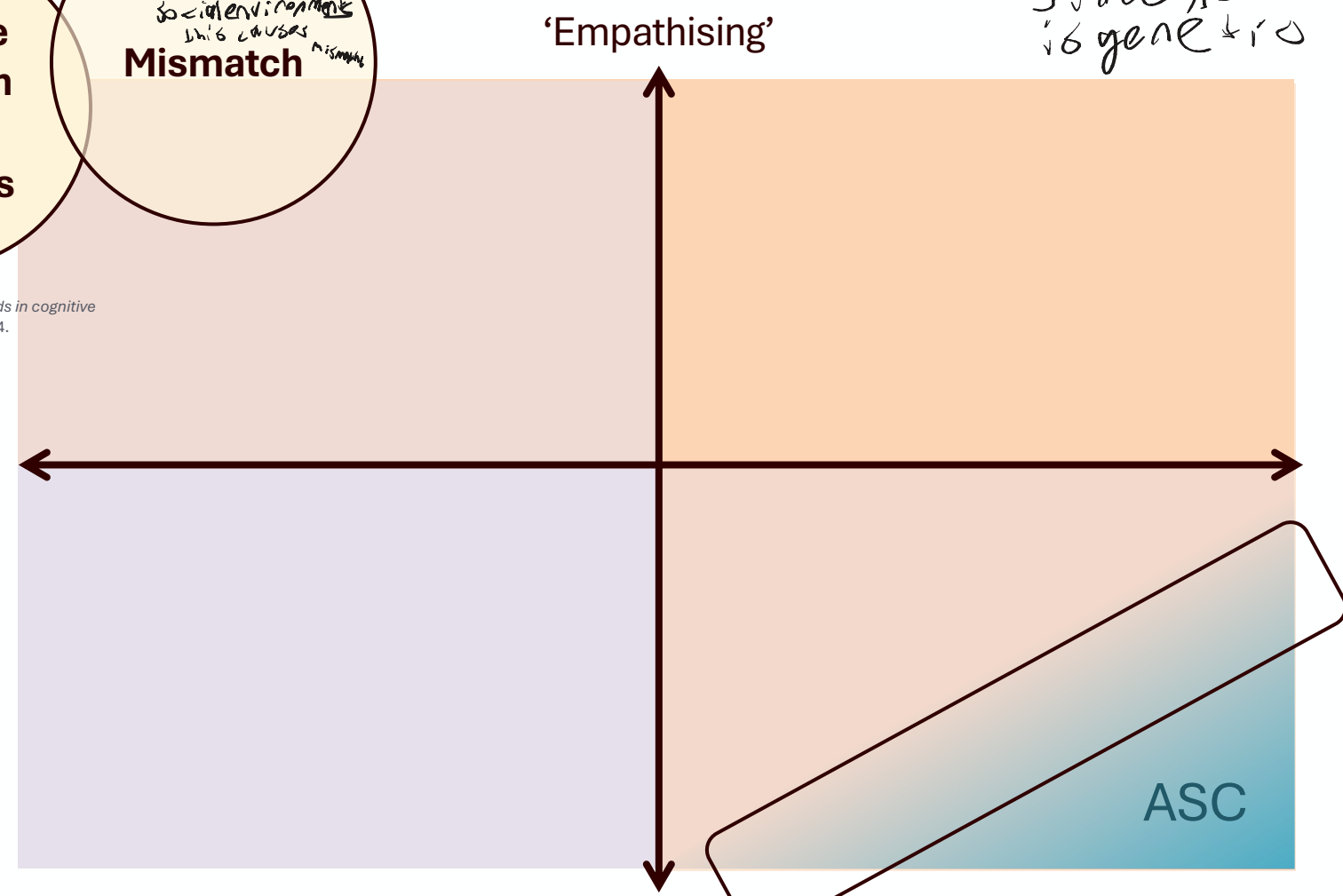
ASD



(Some) evolutionary explanations for ASC



Baron-Cohen, Simon (2002) *Trends in cognitive sciences* 6.6: 248-254.



Handwritten note: "We can think abt this evolutionary since ASD is genetic"

Throwback Thursday

a	Proximate	Ultimate
	Causation	Current Utility
Contemporary		
Historical	Ontogeny	Evolution

'Systemising'



Crespi & Badcock (2008). *Behavioral and Brain Sciences*.

Give it to me straight, doc

1. There is no objective boundary between 'normal' and 'abnormal'.
2. To explain something, you need to know if it qualifies as a trait that selection can operate on.



Depression:

1. Evolutionary explanations often focus on depression as a trait.
2. These explanations focus on why this trait might be adaptive.
3. However, there are many issues with thinking of depression as a trait that was selected for, not least because people with depression tend to have lower fitness!
4. Instead, we might want to think about depression as a descriptive term that describes the state of a system that increases or decreases responsiveness: the mood system.
5. The mood system is, itself, **adaptive**. This means that...
 - Depression can arise as an **adaptive** response to poor conditions (in which case we're not dealing with pathology per se)
 - Depression is **maladaptive** if the mood system went awry

Autism:

6. Autism, like depression, exists at one end of a continuous distribution – this time, the E-S distribution
7. But, unlike depression, an individual's point on this distribution is relatively fixed. Autism is highly heritable. In theory, this makes it easier to ask evolutionary questions about.
8. Evolutionary explanations broadly suggest that autistic traits were selected for in our evolutionary past – that they were **adaptive**.

Schizophrenia

From previous
lecture

- 1 Group living
- 2 Shared ritual
- 3 (In)equality
- 4 Childcare

Schizophrenia is characterized by

↳ complex mental health disorder

- **Psychosis:** a loss of touch with reality; difficulty understanding what is real and what is not
- **Delusions:** fixed false beliefs held despite clear or reasonable evidence that they are not true
 - Most common: **persecutory (or paranoid) delusions**, when a person believes they are being harmed or harassed by another person or group
- **Hallucinations:** hearing, seeing, smelling, tasting, or feeling things that are not there



Difficult identifying what is real in surroundings

Risk is higher in

- Males
- First- and second-generation immigrants
- Urban settings
- Minority populations living in areas with a low own group density

↳ few members of own perceived group

Outgroup intolerance hypothesis

- Mismatch in group composition between ancestral condition and modern condition
- Ancestral condition (in which the human brain evolved):
 - close kinship
 - clear demarcation between ingroup and outgroup membership
 - Kin and ingroup members: care, comfort, support
- Brain expects input from a social environment that has a high density of kin and ingroup members during critical stages of development.
 - ↳ kin & ingroup: you expect them to give you care & comfort.
 - Brain expects environment w/ people like you who care for you
- Modern condition:
 - abundance of outgroup members and/or paucity of ingroup members

Risk is higher in

- Males
- First- and second-generation immigrants
- Urban settings
- Minority populations living in areas with a low own group density

↳ People unlike you → signals of being in outgroup
+ people in outgroup could hurt you or take resources from you

